



GAME THEORY



Game Theory

- Study of strategic decision-making among rational individuals.
- Outcomes depend on your actions and the actions of others.
- Focus: strategies, payoffs, equilibria.
- Applies to economic markets - pricing wars, politics, environmental agreements, elections etc.

Prisoner's Dilemma

The story:

Two suspects are arrested. The police think they were going to rob a store but only have enough evidence to charge them with trespassing.

If they are charged with trespassing, they get 1 months in jail; but for attempted robbery they get 12 months.

The police want to get someone on robbery charges but for that they need one of the criminals to rat the other one out.

As an incentive to be rat, the rat will get to go home free and the quiet one will get a harsh 12 months.

If they both confess, then they both get 8 months as a leniency relative to the full sentence for confessing.

Prisoner's Dilemma

- Interrogator separates them and offers a deal:
 - Confess: Go free if other stays quiet.
 - Both confess: 8 months each.
 - Both stay quiet: 1 month each.
 - Stay quiet while other confesses: 12 months.

Payoff matrix:

		Player 2	
		Quiet	Rat
Player 1	Quiet	$\frac{-1}{P1}, \frac{-1}{P2}$	-12, 0
	Rat	0, -12	-8, -8

Payoff matrix:

		Player 2	
		Quiet	
Player 1	Quiet	-1, -1	
	Rat	<u>0</u> , -12	

Player 2

Quiet

Rat

-12, 0

-8, -8

Dominant Strategy

- ***Definition:** Strategy x strictly dominates strategy y for a player, if x generates a greater payoff than y regardless of what the other players do.*
- For player 1, the rat strategy “Strictly dominates” the quiet strategy - He is better off ratting regardless of player 2's choice
- Putting these two pieces of information together, we reach an important conclusion—player 1 is better off confessing regardless of player 2's strategy!
- Thus, player 1 can effectively ignore whatever he thinks player 2 will do, since confessing gives him less jail time in either scenario. So, **player 1 has the dominant strategy.**

Payoff matrix:

		Player 2	
		Quiet	Rat
Player 1	Quiet	-1, -1	-12, <u>0</u>
	Rat		

Payoff matrix:

		Player 2	
		Quiet	Rat
Player 1	Quiet		
	Rat	0, -12	-8, <u>-8</u>

Dominant Strategy

- Let's switch over to player 2's perspective. As before, **player 2 should confess on both occasions**, as he will spend fewer months in jail months in jail.
- **For player 2, the rat strategy “Strictly dominates” the quiet strategy** - He is better off ratting regardless of player 1's choice.
- Once again, player 2 prefers confessing regardless of what player 1 does. So, **player 2 also has the dominant strategy.**

Payoff matrix:

		Player 2	
		Quiet	Rat
Player 1	Quiet	-1, -1	-12, 0
	Rat	0, -12	-8, -8

Nash Equilibrium

- Thus, we have reached a solution: both players confess, and both players spend eight months in jail.
- Compare the (quiet, quiet) outcome to the (confess, confess) outcome. (quiet, quiet) outcome leaves both players better off than the (confess, confess) outcome.
- As we just saw, promises to remain silent are unsustainable. Player 1 *wants* player 2 to keep quiet so when he confesses he walks away free. The same goes for player 2. As a result, the **(quiet, quiet) outcome is inherently unstable.**
- Ultimately, the players finish in the **inferior (but sustainable) (confess, confess) outcome.** This is what we call **Nash Equilibrium.**

Nash Equilibrium

A “Nash equilibrium” is such a strategy in a game such that no player can benefit by unilaterally altering his decision.

Steps:

- 1) Look for strictly dominated strategies
- 2) Eliminate strictly dominated strategies
- 3) Mark best response for player one for all possible actions of players two
- 4) Mark best response for player two for all possible actions of players one
- 5) Cells where both players have best response marked is your pure strategy Nash equilibrium.

Assumptions:

- humans are rational beings
- humans always seek the best alternative in a set of possible choices

Applications of PD #1:

War and First-Strike Advantage

- First, consider two states considering whether to go to war. The military technology available to these countries gives the side that strikes first a large advantage in the fighting.
- In fact, the first-strike benefit is so great that each country would prefer attacking the other state even if its rival plays a peaceful strategy.
- However, because war destroys property and kills people, both prefer remaining at peace to simultaneously declaring war.

Applications OF PD #1:

War and First-Strike Advantage

	Defend	Attack
Defend	3, 3	1, 4
Attack	4, 1	2, 2

- From this, we can see that the states most prefer attacking while the other one plays defensively. (This is due to the first-strike advantage.)
- Their next best outcome is to maintain the peace through mutual defensive strategies. After that, they prefer declaring war simultaneously. Each state's worst outcome is to choose defense while the other side acts as the aggressor.
- We do not need to solve this game—we already have! This is the same game from the previous section, except we have exchanged the labels “quiet” with “defend” and “confess” with “attack.”
- Thus, we know **that both states attack in this situation even though they both prefer the (defend, defend) outcome.** The first-strike advantages trap the states in a prisoner's dilemma that leads to war.

The Prisoner's Dilemma - example

- Two suspects are arrested and interrogated separately.
- Cooperate (Stay Silent): Trust the other prisoner.
- Defect (Betray): Testify against the other prisoner.
- Their outcomes depend on the combination of choices.

Payoff Matrix

	B: Cooperate	B: Defect
A: Cooperate	(-1, -1) Light sentence.	(-5, 0) A cooperates, B defects.
A: Defect	(0, -5) A defects, B cooperates.	(-3, -3) Both defect: moderate sentence.

Key Takeaways

- Dominant strategy: Defect. Each does better by betraying.
- Nash Equilibrium: Both defecting is stable.
- Rational decisions can lead to worse outcomes for all.

Real-World Example: Cartel Pricing

- Scenario: Two airlines agree to keep ticket prices high.
- Each airline tempted to secretly lower prices.
- If both lower prices, profits decrease.
- Dominant Strategy: Lower prices.
- Nash Equilibrium: Both lower prices.